B.Sc. Engg. 4th Year Odd Semester 2015

DBMS LAB

DISCUSSION-1

Use the Pubs database and explore/perform the following topics/ tasks.

1. Knowing the names of all the tables in a database.

   Discussion

   The sysobjects system table holds data about all the tables and other database objects.

   You can write and execute the following command to see what this system table holds:

   SELECT * FROM SYSOBJECTS

   But only to get the table names, the more specific command is

   SELECT NAME FROM SYSOBJECTS WHERE XTYPE='U'

2. Show all/specific records with all/ specific fields from a table .

   Discussion

   i) To show all the records with all the fields, the command is

   SELECT * FROM <TABLENAME>

   Example: To show all the records from the authors table .

   SELECT * FROM AUTHORS

   ii) To show all the records with specific fields, the command is,

   SELECT <COLUMNNAME1, COLUMNNAME2.> FROM AUTHORS

   Example: To show author last name and state all for authors from the authors table .

   SELECT AU_LNAME, STATE FROM AUTHORS

   iii) To show all fields of specific records satisfying a condition, the command is

   SELECT * FROM AUTHORS WHERE <COLUMNNAME> <CONDITION>

   Example: To show all the fields of those authors who live in the state of CA

   SELECT * FROM AUTHORS WHERE STATE = ''CA'

   Example: To show all the fields of those authors who have a last name `White` and live in the state of CA

   SELECT * FROM AUTHORS WHERE AU_LNAME='White' AND STATE = ''CA'

   Use the Titles table for the following two tasks.

   Task1: Show the name of the books which have yearly total sales of more than 8000.

   Task2: Show the name of the books which have royalty of 12 to 24.

3. Showing ordered list

   Discussion: Showing an ordered list. Titles tables sorted on the price field uppose we want to see the maximum price of the books. The command is

   SELECT * FROM TITLES ORDER BT PRICE ASC

Also try DESC.

4. Showing aggregate values

   Discussion: Suppose we want to see the maximum price of the books. The command is

   SELECT MAX(PRICE) from TITLES

   Similarly FOR AVERAGE price

   SELECT AVG(PRICE) from TITLES

   MAX(), AVG() are SQL functions.

5. Showing aggregate values within groups

   Discussion: Suppose we want to show the book type and the average price of the each type. The command is

   SELECT TYPE , MAX(PRICE) from TITLES GROUP BY TYPE

6. Showing aggregate values within groups having some condition

   Discussion: Suppose we want to show the book type and the average price of the each type if average price is higher than a given values, say, 15. The command is

   SELECT TYPE, AVG(PRICE) from TITLES GROUP BY TYPE HAVING AVG(PRICE) > 15.

   Task3 • Show average price of the books of each type, the total yearly sales of that book type along with book type

7. Showing formatted string with customized header.

   Discussion: We want to show the author name and ph no. where the name should be in the following manner,

   J. White, i.e. First letter of first name dot last name

   The command will be

   SELECT "Name"=SUBSTRING(au_fname,1,1)) + '. '+ au_lname, phone FROM authors

   "Name" is the column heading on the output

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DISCUSSION-2

Use the Pubs database and explore/perform the following topics/ tasks.

1. Joining Tables (Inner join]

   <u>Discussion</u>

   The following query shows author's last name and title id of books.

   SELECT au_lname, title_id FROM authors JOIN titleauthor ON authors.au_id=titleauthor.au_id

   Task 1:

   i) Show the title of a book, the corresponding author name.

   ii) Show the title of a book, the corresponding author and publisher name.

2. The Cartesian product

   The following query shows author's last name and city along with publisher's name and city

   SELECT au_lname, pub_name FROM authors, publishers

   Task 2:

   iii) Show the author name, city, publisher name and city only for which the authors and the publishers live in the same city

3. Nested query

   SELECT * FROM titles WHERE royalty = (SELECT avg(royalty) FROM titles)

   Task 3:

   iv) Show the author name(s) of the book which has the maximum royalty

   Hint: use IN in place of =

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4. Creating a table:

   A table called CustomerAndSuppliers is to be created on the following schema

Field name	Data type	Size	Requirement
cust_id	Character	6	1. Primary key 2. Starting with C or S and then 5 digits i.e. C00001 or S0001.
cust_fname	character	15	NULL not allowed
cust_lname	character	15	Use variable character size
cust_address	text		
cust_telno	character	12	Must follow the format like 012-34567890
cust_city	character	12	Default value is Rajshahi
sales_amnt	money		Negative values not allowed
proc_amnt	money		Negative values not allowed

The corresponding SQL statement is as follows

```
CREATE TABLE CustomerAndSuppliers
(
        cust_id CHAR (6) PRIMARY KEY CHECK (cust_id LIKE '[CS][0-9][0-9][0-9][0-9][0-9]'),
        cust_fname CHAR(15) NOT NULL,
        cust_lname VARCHAR (15),
        cust_address TEXT,
        cust_telno CHAR (12) CHECK (cust_telno LIKE '[0-9][0-9][0-9]-[0-9][0-9][0-9][0-9][0-9][0-9][0-9][0-9]'),
        cust_city CHAR (12) DEFAULT 'Rajshahi',
        sales_amnt      MONEY CHECK (sales_amnt>=0),
        proc_amnt       MONEY CHECK (proc_amnt>=0)
)
```

5. Inserting data into a table:

Example:

```
INSERT CustomerAndSuppliers
(cust_id,cust_fname,cust_lname,cust_address,cust_telno,cust_city,sales_amnt,proc_amnt) VALUES
('C00001','Iqbal','Hossain','221/B Dhanmondi','017-00000000','Dhaka',0,0)
```

Task 4: *Create tables on the following schema*

Table name: Item

Field name	Data type	Size	Requirement
item_id	Character	6	1. Primary key 2. Starting with P and then 5 numbers i.e. A0001.
item _name	character	12	
item_category	character	10	Example: Electrical, Mechanical, Software, Books etc.
item _price	float	12	Negative values not allowed
item_qoh	integer		Negative values not allowed
item_last_sold	date		Default value is current date

Table: Transactions

Field name	Data type	Size	Requirement
tran_id	Character	10	1. Primary key 2. Starting with T and then 9 numbers i.e. T000000001.
item _id	character	6	Foreign key with reference to item table
cust _id	character	5	Foreign key with reference to customer table
tran_type	character	1	Either S or O, (S for sales Order to supplier)
tran_quantity	integer	12	Only positive values.
tran_date	date and time		Default is current date

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DISCUSSION-4

1. Trigger

Trigger is special type of Stored Procedure which is attached to a table and is only executed (also known as fired) when an INSERT, UPDATE or DELETE (i.e., modification of table data) occurs on that table. . One has to specify the modification action(s) that fire the trigger when it is created.

Unlike stored procedures, triggers can not be explicitly executed.

Example: 1 The following trigger shows a message when a row is inserted in Items table

```
CREATE TRIGGER trg_test ON Items FOR INSERT
AS
BEGIN
    PRINT 'Data inserted in Item Table'
END
```

2. Triggers can be used for automatically updating a table when an insert/update/ delete statement takes place in another table. For example, whenever a Transactions takes place, i.e., a row is inserted in the Transactions Table, item_qoh field in the Items table should be updated either by increasing it or decreasing.

In such cases, a table called INSERTED is used by the trigger.

- INSERTED is a temporary table that SQL server automatically creates whenever an insertion takes place in a table. The inserted table holds only the row that is getting inserted.
- Similarly, a temporary table called DELETED consisting of row that is getting deleted is created when a deletion takes place.
- Both INSERTED and DELETED tables get created when an UPDATE statement takes place.

The following trigger updates items table whenever a transaction takes place.

```
CREATE TRIGGER trg_update_item ON Transactions FOR INSERT
AS
BEGIN
    DECLARE @item_id char(6), @tranamount int, @tran_type char(1)
    SELECT @item_id=item_id, @tranamount=tran_quantity, @tran_type=tran_type FROM INSERTED
    IF  (@tran_type ='S')
      UPDATE Items SET item_qoh=item_qoh-  @tranamount WHERE item_id=@item_id
      ELSE
    UPDATE Items SET item_qoh=item_qoh+  @tranamount WHERE item_id=@item_id
END
```

Assignment

Task 1: Write a trigger on Transaction that automatically updates the sold_amnt or proc_amnt field of CustomersAndSuppliers table whenever a transaction happens.